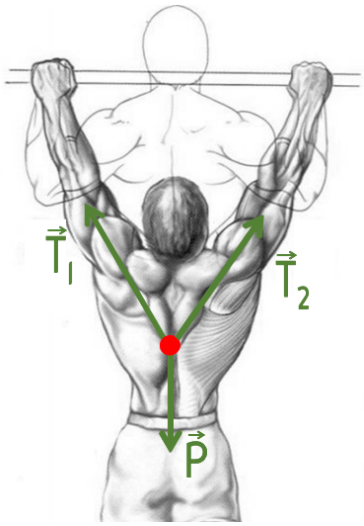
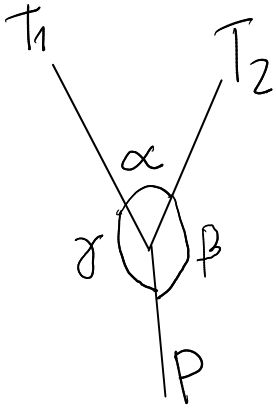
PROFESSOR FERRETTO

Equilíbrio de um Ponto Material – Parte 02

TEOREMA DE LAMY (SOMENTE PARA 3 FORÇAS)

$F_R = 0$





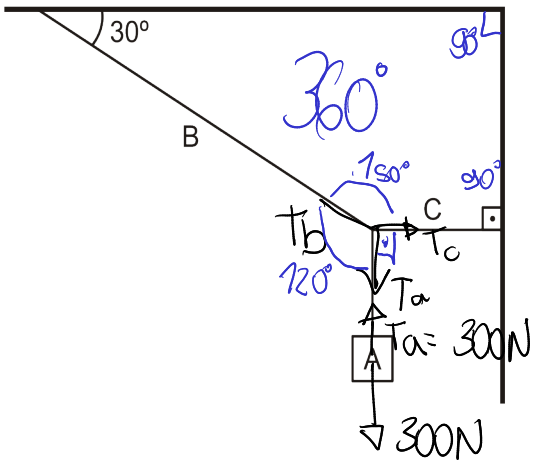
$$\frac{P}{\text{Sen}\,\alpha} = \frac{T_1}{\text{Seno}\,\beta} = \frac{T_2}{\text{Sen}\,\gamma}$$

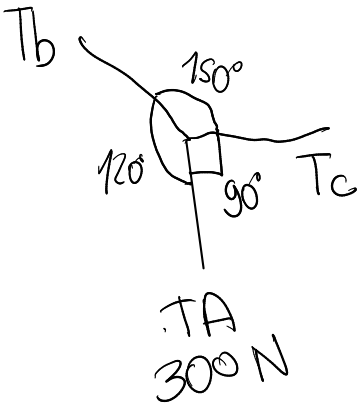
Exercício

(Ifsul) Uma caixa A, de peso igual a 300 N, é suspensa por duas cordas B e C conforme a figura abaixo. O valor da tração na corda B é igual a:

a) 150,0 N.
b) 259,8 N.
c) 346,4 N.
d) 600,0 N.

$T_a = 300N$
$T_b = ?$
$T_c = ?$



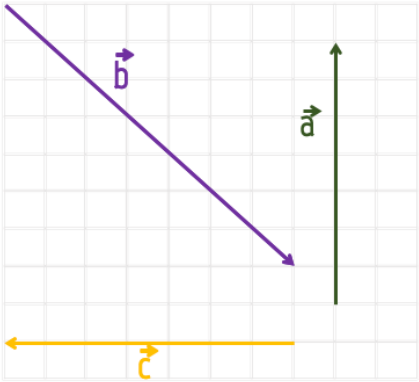


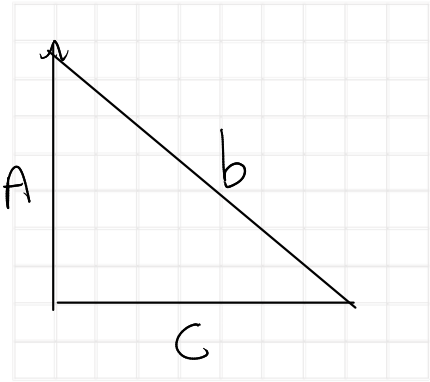
$$\frac{T_b}{\text{Sen}\,90^\circ} = \frac{T_a}{\text{Sen}\,150^\circ}$$

$$\frac{T_b}{1} = \frac{300}{1/2} = T_b = \frac{300}{1/2} = \frac{300}{1} \cdot \frac{2}{1} = 600$$

MÉTODO DO POLÍGONO

$F_R = 0$





$F_R = 0$ //

Exercício

(Ifsul) Uma caixa A, de peso igual a 300 N, é suspensa por duas cordas B e C conforme a figura abaixo. O valor da tração na corda B é igual a:

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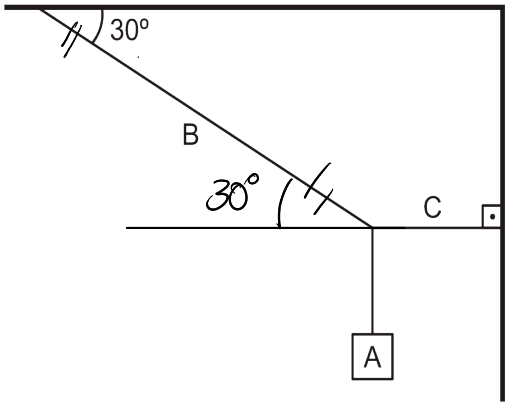
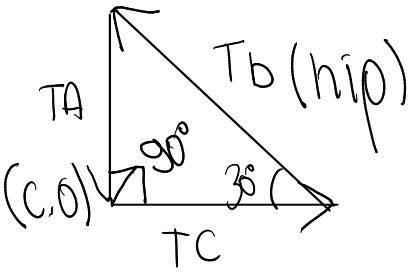
c) 346,4 N.
d) 600,0 N.

$T_a = 300\,N$

$T_b = ?$

$T_c = ?$





$\text{Sen } 30° = \frac{CO}{h}$

$\text{Sen } 30° = \frac{TA}{Tb}$

$\frac{1}{2} = \frac{300}{Tb} = \frac{600}{1} = 600$ //

Anotações: